IoT Based Smart Biometric Locker

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***Abstract*— In our residences, workplaces, retail establishments, and industrial organizations, security was and is a concern. Everyone worries about someone breaking into their house or workplace without their permission and taking valuables out of cabinets or lockers. Locks that require a key to operate can be installed on a typical locker door to allow keyless entry. Alternatives to this system include the locks' password or pattern systems, which run the risk of disclosing the password. A biometric verification system can therefore offer a solution to such problems. Any technique for verifying a person's identity using one or more distinguishing biological traits is referred to as biometric verification. Because each person's fingerprints are distinct, we will use them for biometric verification. Additionally, using fingerprints as a locker key can significantly lessen the security issue of unauthorized individuals accessing our valuables. Furthermore, this system won't lead to problems like key loss since we won't need to carry keys. The locker has an IoT platform that records all data of the user who accesses the locker in addition to enhancing security. This allows the administrator to keep an eye on all data on the server. The workplace can use this project in a variety of ways. To track who and when used a particular facility, can also be installed indoors. Because only authorized users can access the system, it is more secure.**

***Keywords*— *IoT, Biometric identity, authentication, Xamp*.**

I. INTRODUCTION

New, advanced locking system techniques are replacing the conventional mechanical lock and key mechanisms. These methods combine mechanical and electronic devices in highly intelligent ways. These modern lock systems' simplicity and effectiveness are two of their most notable features. This kind of automatic lock system consists of an electronic control assembly that uses a fingerprint to control the output load. The shortcomings, abuse, and numerous flaws of conventional RFID technology are numerous. Therefore, using biometric fingerprint technology is much more reliable and secure. In offices and schools, biometric technology is also used to monitor attendance. To access the locker, this project uses fingerprint authentication; all logs are kept on the server. The people who access the locker can be tracked with the aid of this log. Some users' access to the locker may also be restricted by the server. This technology can be used in smart door systems in addition to smart lockers.

II. RELATED WORK

In contrast to the conventional key-and-lock system, a different system was used in the previous project to increase security. The door-locking system includes a GSM module and a fingerprint sensor. The individual must pass the fingerprint test in order to be authorized. Once the test has been authorized, a second security measure, such as an OTP or an image password, will be selected at random. If the password is an image, that is acceptable, but if the password is an OTP, the user must have a mobile phone on them in order to enter the system because the lock does not offer any options for authentication. If the OTP is not entered, the system will return to the initial stage or first screen. It has a significant drawback in that not everyone carries a mobile phone, and this level of security may irritate users because, frequently, because of the range, even authorized users cannot enter.[1]

A biometric fingerprint sensor is used in another project for authentication. In high-security areas, an optical biometric system is used. However, the project's main drawback is that it won't be as secure if it only uses biometric fingerprint sensors. The authorized person's fingerprint could be stolen and accessed by someone else. If it only has a fingerprint, it can enter a room with authorization and avoid detection, which could be disastrous.[2]

The alternative project made use of an optical fingerprint sensor that communicated via UART with a PIC microcontroller. It serves as the door lock's biometric security system. If the user authorizes, the door will open immediately after the user places their finger on the sensor; otherwise, the door will stay closed.[3] The issue with this project is that using just the fingerprint module is not very secure; therefore, lockers, which require high levels of security, cannot use this kind of system. The admin can easily track user activity so he can quickly identify the user if something goes wrong, which solves all the issues mentioned in the previous system.

III. METHODOLOGIES

A. Bicycle

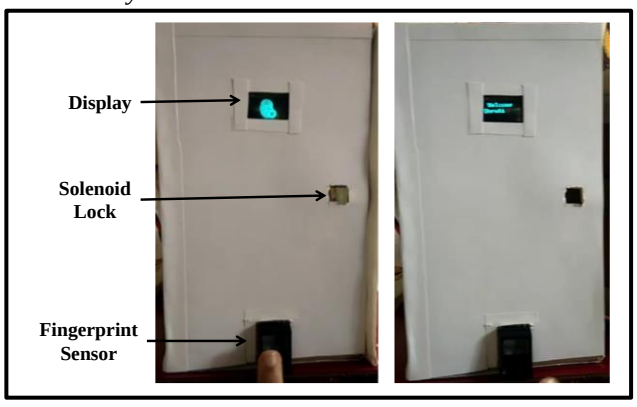


Fig.1. Overview of the project

Fig. 1 depicts the project's appearance and the placement of the modules. The OLED display, which is located on the top, shows all the updates occurring in the locker. The fingerprint sensor is positioned at the base of the locker for finger placement convenience. The door will open and the display will display the name of the valid user whenever the finger is placed on the sensor on the left side of the solenoid lock.

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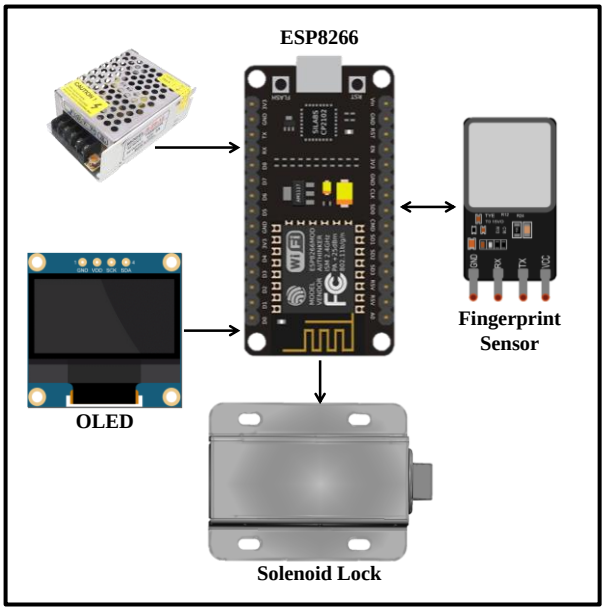


Fig.2. Block diagram of IoT-based smart biometric Locker

This project will be stationary, so it uses a switching mode power supply (SMPS). because it can supply consistent DC voltage from the AC mains. Due to the fact that some devices only need 3.3 volts to operate while others need 12 volts, the voltage output from the SMPS must be regulated at various voltage levels. This project needs to be constantly connected to the internet in order to upload the data to the server. Wi-Fi modules are the best way to connect to the internet without using a lot of power, so AI thinker NodeMCU ESP8266 is used as it has built-in Wi-Fi support. This also functions as a microcontroller to manage every other gadget. It is compatible with a variety of peripheral interfaces, including GPIO, PWM, I2C, and SPI.[6]

The display is used to show what is happening inside the processor. OLED 0.96-inch display is the one being used. Compared to a standard 16x2 LCD, it has a more attractive user interface thanks to its 128x64 dot matrix resolution. The synchronous serial protocol known as the I2C protocol is used by this module to communicate with the microprocessor. Regardless of whether the connection has been made, this display indicates the Wi-Fi status. Additionally, it indicates whether the fingerprint is valid or invalid; if it is valid, the name of that specific person will be displayed. Because it indicates when to scan the finger, this display is crucial.[4]

This project's most crucial module is the fingerprint one. It is an optical fingerprint sensor that relies on the phenomenon of total internal reflection. When there isn't a finger on the prism, all of the light will be completely reflected off of it, leaving the image sensor with a simple image. The intensity of refraction changes as soon as the finger is placed on the prism due to a change in the refractive index, and the image sensor notices this change. This module uses the asynchronous UART protocol to communicate with the microprocessor. The fact that this module has internal memory and can store 1000 fingerprints makes it important. On the module, all of the fingerprint data is kept. It will compare the most recent fingerprint data with the stored data while scanning the finger. If a match is made, the module will communicate the microprocessor's unique ID for the corresponding fingerprint. The user's data will be displayed on the screen after the microprocessor processes the ID.[5]

The solenoid lock is an electronically controlled lock that retracts the latch bolt when supplied with 12 volts. As soon as the current is cut off, the latch bolt returns to its initial position. When the solenoid coil in the lock receives current, it turns into an electromagnet and pulls the latch bolt in the direction of the lock. Due to the voltage difference, the control signals from the microprocessor cannot be sent directly to the lock; therefore, additional circuits are needed to convert this signal to a higher voltage. This circuit uses a transistor to drive a relay, which is used to isolate the 12-volt and 5-volt sides because they can harm other devices.

B. Server

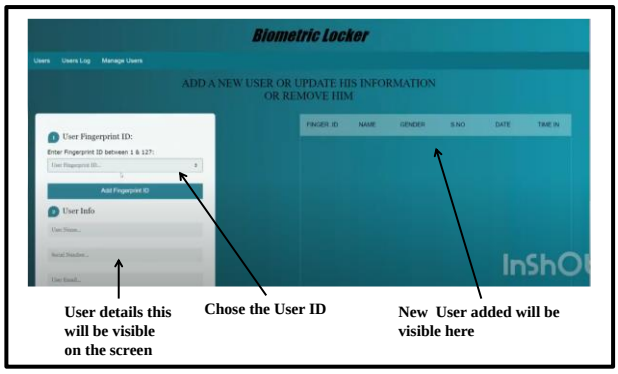


Fig.3. New user screen

The local server was made using the XAMP software, which was also used to create the server. The locker uses Wi-Fi to connect to this local server. The server page to add a new user to the locker is shown in Figure 3. The administrator will choose a user ID number between 1 and 127, which is specific to each user when adding a new user to the locker. The fingerprint that corresponds to this number will be stored and sent to the fingerprint scanner. This number will be transmitted by the fingerprint to the microcontroller as the user's identity each time this user tries to open the locker. Now that the user has chosen a number, the administrator will click Add Fingerprint ID to add the user's information to the list on the left. Additionally, the user must scan his fingerprint on the locker before the server is notified. After scanning is complete, the administrator must add all user-related information, including the user's name, serial number, email address, etc. Therefore, the user's information will be visible on the screen and even on the server whenever he or she accesses the locker.

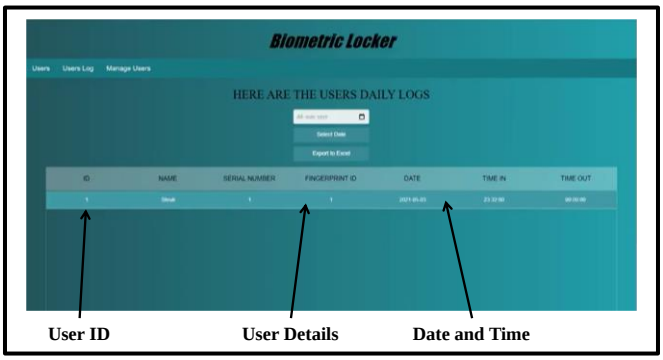


Fig.4. User logs

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As seen in figure 4, the server will record the user who accessed the locker. The database will store the user's ID number, name, serial number, fingerprint ID, date, and time. This information can be exported to an excel file. The server keeps date-based entries for past records, which the admin can also view. The administrator can check the entry data to determine who must have taken something if it is missing from the locker.

C. Procedure

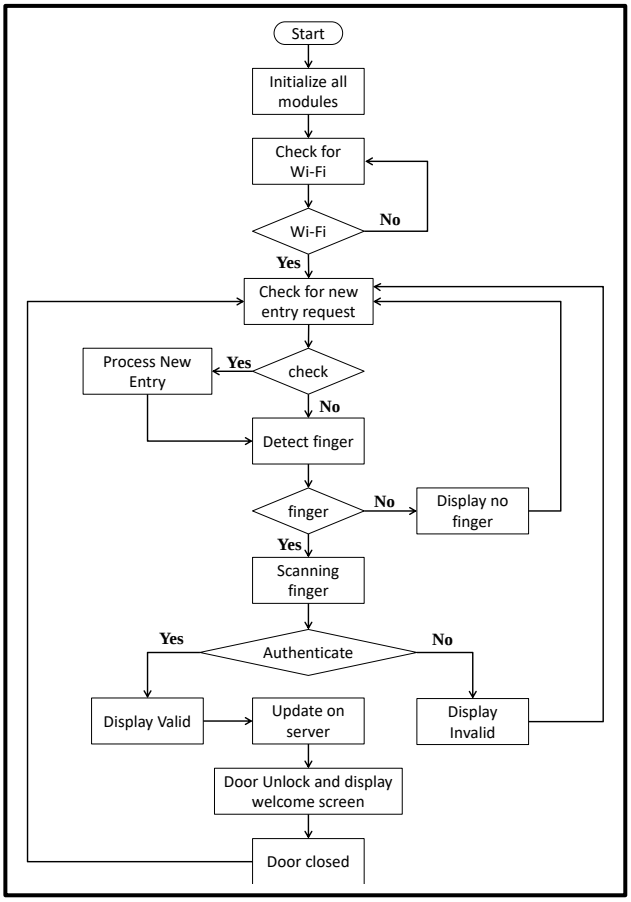


Fig.5. Flow chart of overall working

The process begins with the initialization of all modules. During this step, the OLED display is started, and an image is set to show that it is operational. A fingerprint sensor will then be initialized and placed in scanning mode after that. The Wi-Fi module will begin connecting once all of the modules have been initialized; the Wi-Fi credential will already be loaded into the microcontroller. The microcontroller will attempt to connect to the Wi-Fi, but the process will not advance until it is connected. Due to the importance of the network connection, it is an infinite loop. The process will continue once the Wi-Fi is established.

Locker will now recognize any requests to add new users or remove existing users. The microcontroller will activate the record mode on the fingerprint sensor if a request to add a new user is received, allowing the user's fingerprint to be captured and stored on the sensor. The fingerprint sensor will return to scanning mode after this request is fulfilled so it can authenticate the user's fingerprint.

The fingerprint sensor only activates for a short amount of time before turning off. It first checks to see if any finger is pressed against the sensor; only then does it begin to scan the fingerprint. Using a digital signal processor, the scanned fingerprint data is compared to the stored fingerprint data (DSP). If a match is made, the processor will receive the user ID for the print that matched. When an ID is received, the processor will recognize it as a legitimate user. In accordance with the ID that was received, the microcontroller will request data from the server and display information on the screen. The sensor will return NULL to the microcontroller if the fingerprint does not match the user who is authorized to open the locker. After receiving NULL, the microcontroller will attempt to scan a different fingerprint because it will be known that the user is invalid and will be indicated on the screen.

The locker will update the time on the server when it is unlocked, and when the user closes it, the microcontroller will know that it has been closed. As a result, the microcontroller will update the out time on the server. This is accomplished by placing a switch at the door. When the door is closed, it will press the switch, alerting the microcontroller. This is crucial because the administrator needs to know how long the locker was open.

IV. Observations

Table I. Observation Table

Observed Parameters	Seconds
Overall Starting Time	8
Fingerprint scan time	3
Server Update Time	5

This observation was made after numerous test runs. The module initializing and connecting to the Wi-Fi router is included in the overall startup time. From the moment the finger was placed on the scanner until the display was updated, the fingerprint scan time was recorded. The server update time refers to the amount of time needed to upload all of the data from the device to the server or the interval between when a change is made on the device and when the server is updated.

V. Results

The majority of the project's test runs went according to plan. However, after repeated uses, the sensor was unable to detect properly because of the oil and dirt that stick to the finger and adhere to the sensor. The dirt on the sensor was obstructing this light because the sensor relies on light to function. As a result, for the sensor to function properly after repeated use, it must be clean.

VI. Discussion

The previous project that used a biometric fingerprint system for security has one or more flaws. Our project can solve all the issues and even make the lockers more secure. This project can be applied to any system that needs authorization for entry, not just lockers. In addition to this project, security can be improved by using cameras, password protection, etc.

Future Direction

- Boost security by including a backup power supply so that even if the locker is attempted to be unplugged, it will continue to function. Additionally, it can have the ability to notify the admin via an emergency message that the locker has been unplugged.

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- The camera will begin recording whenever the locker is opened or accessed, which means the user's face will be captured.

- Use GSM to notify the admin if someone is trying to open the locker forcefully. The admin may get an emergency message so he or she can take preventive measures.

CONCLUSION

The Biometric IOT locker system operates flawlessly and according to plan. It completed all of its tasks without any glitches or delays. The sensor must always be kept clean because it is sensitive to dirt and will stop working otherwise. This project improves security systems to which only specific authorities have access. It is preferable to use it because it offers a better solution than using a key.

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